

Nitte (Deemed to be University)

(Established under Sec. (3) of UGC ACT, 1956)

First Semester M.Tech (CBCS) Degree Examinations, December 2025

25CST123 – COMPUTATIONAL STRUCTURAL MECHANICS

(For CST)

Duration: 3 Hours]

[Max. Marks: 100

Note: Answer **Five full** questions choosing **Two full** questions from **Unit-I** and **Unit-II** and **One full** question from **Unit-III**.

Unit – I

Marks BT* CO* PO*

1. a) Define static indeterminacy and find the static indeterminacy of structures shown in figure. Q.1(a).

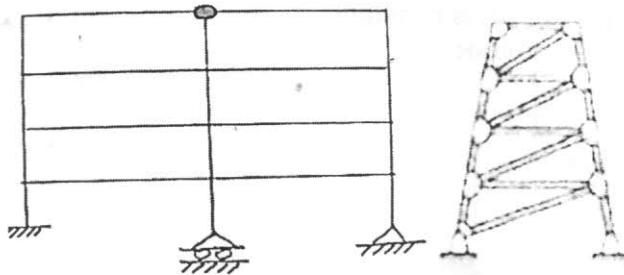


Figure. Q.1(a)

10 L2 1 1,2

- b) Generate the flexibility matrix for the given coordinates of the structure as shown in figure. Q.1(b).

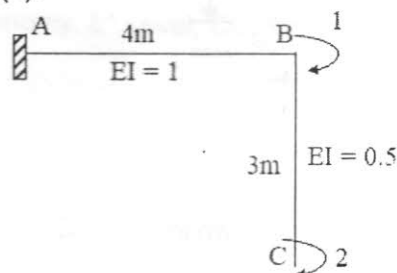


Figure. Q.1(b)

10 L3 2 1,2

2. a) Define kinematic indeterminacy and find the kinematic indeterminacy of structures shown in fig. Q.2(a).

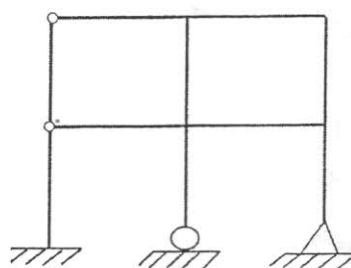
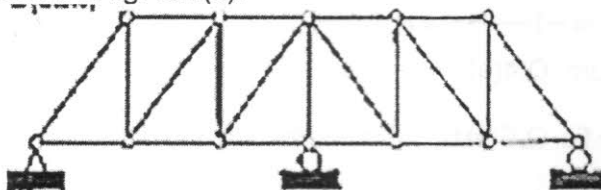


Figure. Q.2(a)

5+5 L2 1 1,2

- b) Generate the stiffness matrix for the given coordinates shown in figure. Q.2(b).

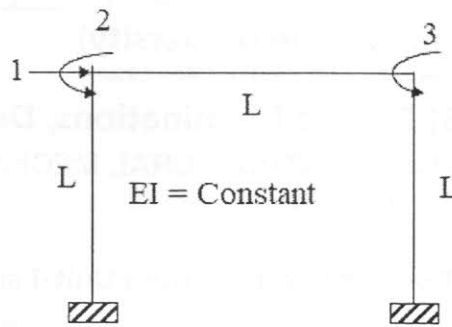


Figure. Q.2 (b).

10 L3 2 1,2

3. Using matrix stiffness method, analyze the truss for the member forces in the truss loaded as shown in figure. Q.3(a). Take AE is constant.

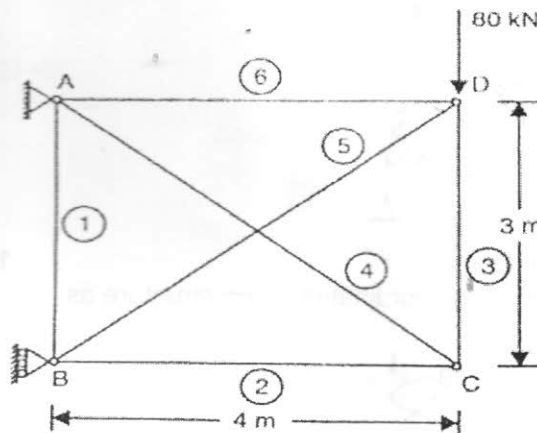


Figure. Q.3(a)

20 L4 2 1,2

Unit – II

4. Analyze the continuous beam by Stiffness method shown in figure. Q.4(a). Draw BMD. Take EI is constant.

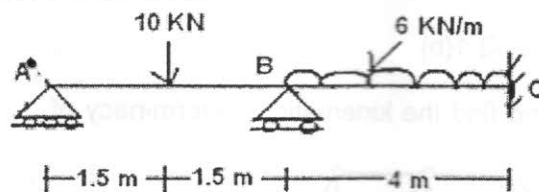


Figure. Q.4(a)

20 L4 3 1,2

5. Analyze the portal frame shown in fig. Q.5(a) by stiffness method and draw BMD.

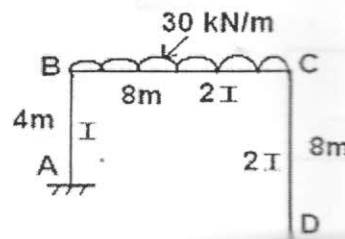


Figure. Q.5(a)

20 L4 3 1,2

6. Analyze the continuous beam shown in figure. Q.6(a) by direct stiffness method. Draw BMD. Take EI is constant.

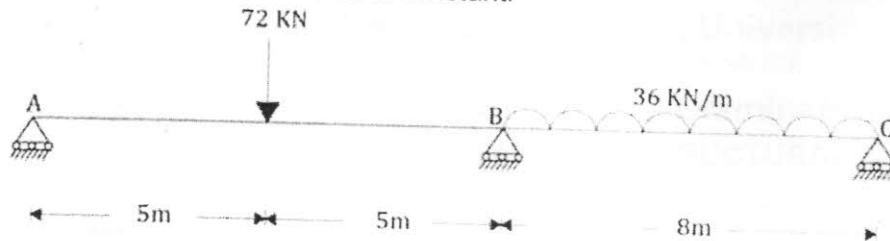


Figure. Q.6(a)

20 L4 4 1,2

Unit – III

7. Solve the following system of equations by Cholesky method.

$$4x_1 + 2x_2 + 14x_3 = 14$$

$$2x_1 + 17x_2 - 5x_3 = -101$$

$$14x_1 - 5x_2 + 83x_3 = 155$$

10+10 L3 5 1,2

8. Solve the following system of equations by Gaussian elimination method.

$$2x_1 - 3x_2 + 4x_3 = 2$$

$$4x_1 + x_2 + 2x_3 = 2$$

$$x_1 - x_2 + 3x_3 = 3$$

20 L3 5 1,2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
